Worksheet: Stripping column separations

Name(s) ________________________________

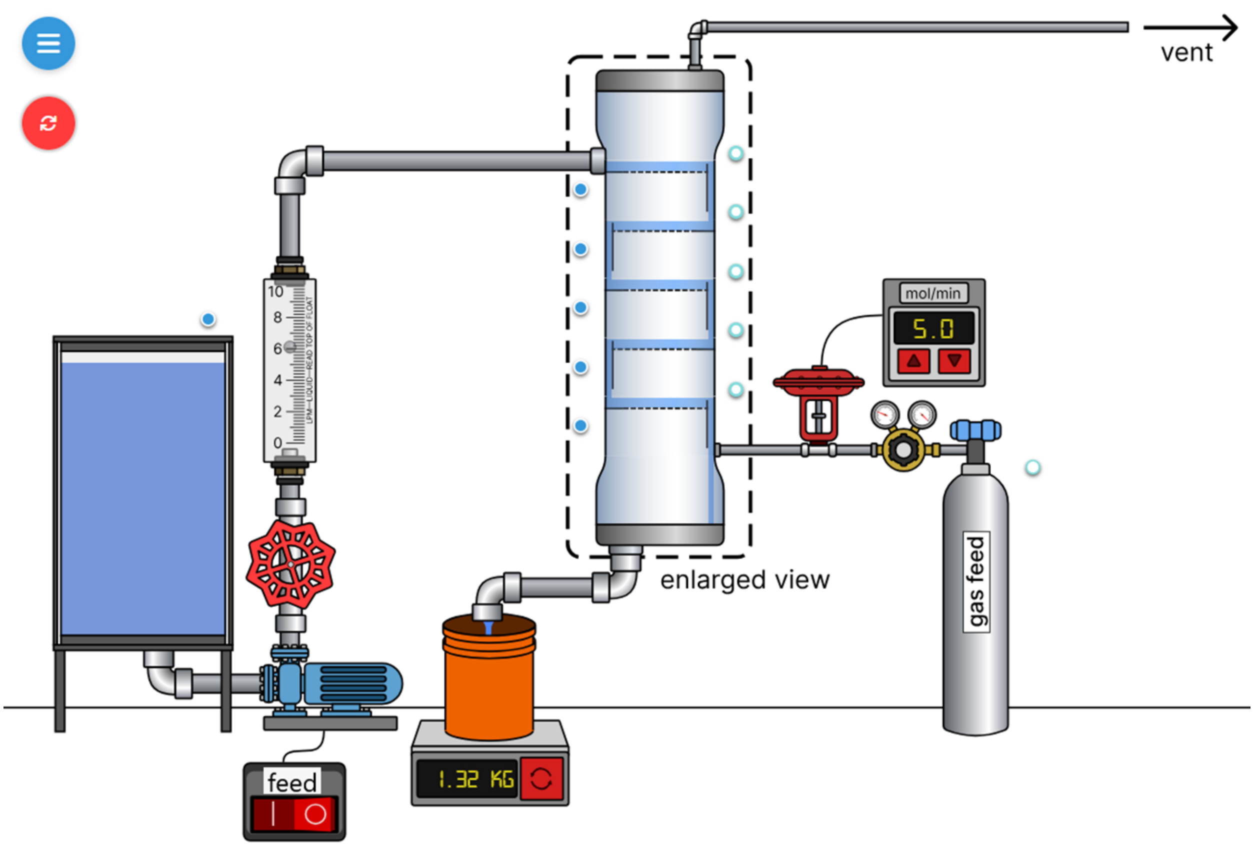
This experiment uses a trayed stripping column that removes a dilute solute from a downward-flowing liquid feed by transferring the solute into an upward-flowing gas stream. The column operates at steady-state at constant pressure. This experiment investigates how the following affect stripping performance:

- number of stages
- stage efficiency
- system pressure
- liquid and gas flow rates

Student learning objectives

1. Be able to perform solute material balances on a column.
2. Apply Henry's law to relate gas and liquid compositions.
3. Determine and be able to explain the effects of pressure, relative flow rates, number of stages, and stage efficiency on stripping column separations.
4. Be able to construct and interpret equilibrium and operating lines on an y versus x diagram.

Equipment



The system consists of a series of equilibrium stages where the liquid phase (water) flows down and the gas phase flows up. The phases leaving each stage are in phase equilibrium. Because the impurity solute concentrations are low, the liquid and gas flow rates are approximately constant from stage to stage. For dilute systems, Henry's law ($y_i P = H x_i$) describes the phase equilibrium.

Notation

x_i = solute mole ratio in liquid leaving stage *i*

y_i = solute mole ratio in gas leaving stage *i*

L = liquid molar flow rate

V = gas molar flow rate

P = absolute pressure

H = Henry's constant (bar)

Note that the solute mole ratios are in ppm.

Questions to answer before starting the experiment

1. How will increasing the number of stages affect the separation? Why?

2. How will decreasing stage efficiency affect the separation? Why?

3. If the gas flow rate increases, how is the separation affected?

4. How does higher pressure affect stripping performance? Why?

5. What is the advantage of counter-current flow over co-current flow?

Experimental Procedure:

Experiment #1 Base case

Use a stripping column with three stages and 100% stage efficiency to measure flow rates and compositions at one pressure. An overall solute material balance and the percent solute removed are calculated.

1. Use the menu on the top left to edit internals. Select the number of stages as 3 and the stage efficiency as 100%.
2. Switch on the liquid feed pump (left) to fill the column. Then, drag and rotate the red feed valve to change the liquid flow rate to a value that is in the middle of the rotameter range.
3. After liquid appears at the column outlet, the gas cylinder valve turns blue. Then click on the blue valve to open it and pressurize the column. Rotate the gas regulator valve to select a gauge pressure of 3.0 bar. It may be easier to adjust the pressure regulator by zooming with the scroll wheel. Use the buttons on the flow controller to adjust the gas flow rate.
4. Change the liquid and/or gas flow rates so that the gas removes about 80% the solute from the liquid feed.
5. Record the inlet liquid volumetric flow rate in Table 1. Calculate the mass flow rate and the molar flow rate (the liquid is water) and enter the values in Table 1.
6. Allow the system to reach steady state. View liquid and vapor solute mole ratios by hovering over the blue dots (liquid) and white dots (gas) next to the column. Assume steady-state if the solute mole ratios change by less than 2% in 30 s.

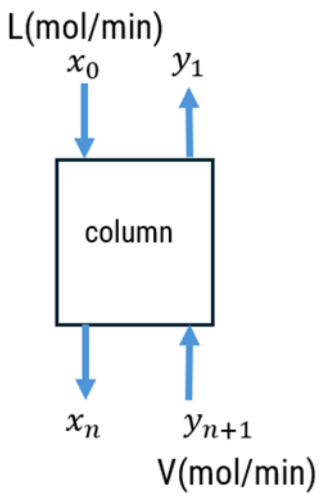
Table 1 Liquid and gas flow rates

Liquid			Gas
Volumetric flow rate (L/min)	Mass flow rate (g/min)	L(mol/min)	V(mol/min)

7. Move the mouse over the buttons above the liquid and gas feeds and record their solute mole ratios in Table 2. Likewise, record the solute mole ratios at the exit of the column in Table 2. The figure below shows what the subscripts refer to in the column.

Table 2 Solute mole ratios

Stage	Solute mole ratio in liquid (ppm)	Solute mole ratio in gas (ppm)
Top of column	x_0 =	y_1 =
Bottom of column	x_n =	y_{n+1} =



where L = liquid molar flow rate

V = gas molar flow rate

Data Analysis:

8. Do a solute mass balance for the column in Table 3. Note that ppm molar flow rates are equal to μmol /min.

Table 3 Solute mole balance

Solute in (μmol /min)			Solute out (μmol /min)			Difference	
Liquid	Gas	Total	Liquid	Gas	Total	(In – out)	% difference
x_0L	y_4V	x_0L+y_4V	x_3L	y_1V	x_3L+y_4V		

9. The streams leaving each stage (e.g., x_2 and y_2 for stage 2) are in equilibrium. Record the mole ratios for the three stages in Table 4. Also record the system absolute pressure. Use the data to determine Henry's law constant for each stage from the equation $y_iP = Hx_i$ and enter the values in Table 4. Calculate an average H value.

Table 4 Solute mole ratios and Henry's law constant

Pressure ______ bar absolute

Stage	Solute mole ratio in liquid (ppm)	Solute mole ratio in gas (ppm)	H(bar)
1	x_1 =	y_1 =	
2	x_2 =	y_2 =	
3	x_3 =	y_3 =	
average			

10. Calculate the percentage of solute removed from the liquid feed.

Experiment #2 Number of stages

Some of the same measurements and calculations performed in Experiment #1 are repeated for a column with eight stages but for the same flow rates and pressure.

Do more stages increase or decrease the percentage of solute removed from the liquid?

1. Use the menu on the top left to edit internals. Select the number of stages as 8 and the stage efficiency as 100%.
2. Switch on the liquid feed pump (left) to fill the column. Then, drag and rotate the red feed valve to change the liquid flow rate to the same value used in Experiment #1.
3. After liquid appears at the column outlet, the gas cylinder valve turns blue. Then click on the blue valve to open it and pressurize the column. Rotate the gas regulator valve to select a gauge pressure of 3.0 bar. It may be easier to adjust the pressure regulator by zooming with the scroll wheel. Use the buttons on the flow controller to adjust the gas flow rate to the same value used in Experiment #1.
4. Record the inlet volumetric flow rate in Table 5, then calculate the mass flow rate and the molar flow rate (the liquid is water) and enter the values in Table 5. Enter the gas molar flow rate in Table 5.

Table 5 Liquid and gas flow rates

Liquid volumetric flow rate (L/min)	L(g/min)	L(mol/min)	V(mol/min)

5. Allow the system to reach steady state. View liquid and vapor solute mole ratios by hovering over the blue dots (liquid) and white dots (gas) next to the column. Assume steady state if the solute mole ratios change by less than 2% in 30 s.
6. Move the mouse over the buttons above the liquid and gas feeds and record their solute mole ratios in Table 6. Likewise, record the solute mole ratios at the exit of the column in Table 6.

Table 6 Solute mole ratios

Stage	Solute mole ratio in liquid (ppm)	Solute mole ratio in gas (ppm)
Top of column	x_0 =	y_1 =
Bottom of column	x_n =	y_{n+1} =

7. Calculate the percentage of solute removed from the liquid feed and compare the value to that obtained in a column with 3 stages. Why does increasing the number of stages improve solute removal from the liquid?

   Keep the current settings for the next experiment.

Experiment #3 Equilibrium and operating lines

Use the same setting for eight stages in Experiment #2 to create a y vs. x plot. Plot the Henry's law equilibrium line and the material balance operating line and step off stages.

1. Add the solute mole fractions and the absolute pressure to Table 7.

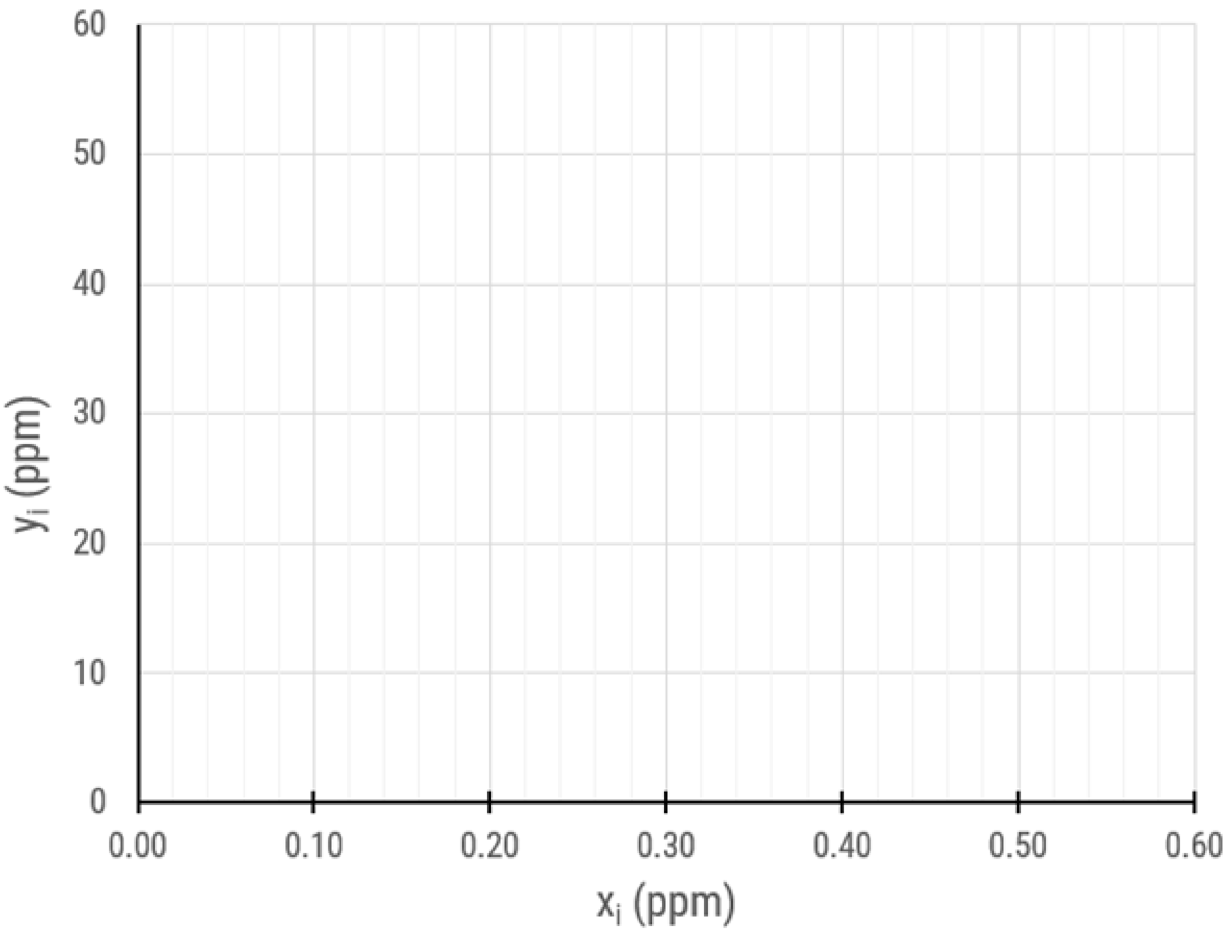
Table 7 Solute mole ratios Pressure ________ bar absolute

Stage	Solute mole ratio in liquid (ppm)	Solute mole ratio in gas (ppm)
Liquid feed	x_0 =	
1	x_1 =	y_1 =
2	x_2 =	y_2 =
3	x_3 =	y_3 =
4	x_4 =	y_4 =
5	x_5 =	y_5 =
6	x_6 =	y_6 =
7	x_7 =	y_7 =
8	x_8 =	y_8 =
Gas feed		y_9 =

2. Plot the equilibrium line ($y_i = \frac{H}{P} x_i$) on the y_i vs. x_i plot below or on a spreadsheet, using the H value obtained in Table 4.
3. Use the liquid and gas flow rates from Table 6 to plot the operating line, which is given by the equation:

$$y_{n+1} = \frac{L}{V} x_n + (y_1 - \frac{L}{V} x_0)$$

4. Count off stages starting at x_0, y_1 on the operating line, then draw a horizontal line to the equilibrium line (x_1, y_1). Then draw a vertical line down to the operating line and repeat until x_8, y_8 .



Experiment #4 Effect of pressure

Do you expect the percent solute removal is higher or lower at higher pressures?

Use the same conditions as in Experiment 2 but change the gas pressure to the values indicated in Table 8, let the system reach steady state, and record the liquid composition leaving the last stage (x_8) in Table 8. Calculate the percentage of solute removed at each pressure.

Table 8 Effect of pressure on solute removal

Absolute pressure (bar)	x_8	% solute removed
2.0		
4.0		
6.0		
8.0		

Why does increasing pressure change the stripping effectiveness for a system that obeys Henry's law?

Experiment #5 Effect of stage efficiency

Use the same conditions as in Experiment 2 but use stage efficiencies of 85% and 65%, let the system reach steady state, and record the liquid composition leaving the last stage (x_8) in Table 9. Calculate the percentage of solute removed for each stage efficiency.

Table 9 Effect of stage efficiency on solute removal

Stage efficiency	x_8	% solute removed
100%		
85%		
65%		

Why does lower stage efficiency reduce separation even when the number of stages is the same?

Questions to answer

1. Where might these measurements have errors?

2. What safety precautions would you take to conduct this experiment in the laboratory?

3. Would separations be more effective at low or high temperature? Why?

4. If the L/V ratio increases, does the percent removal increase or decrease if the number of stages does not change? Explain.

5. Which variable had the strongest effect on the percent removal in this experiment?